

① REQUIREMENTS — canonical enterprise-Q&A RAG

~2,000 users · ~10/40 rps · 9.2K-in / 300-out · quality ~70B



② CONSTRAINTS — three hard gates

MEMORY · LATENCY · ECONOMICS



③ GENERATE CANDIDATES

(a) 8×H100 + prefix cache · (b) P/D 2-pool · (c) KV-quant 7B



CANDIDATE	MEM	LAT	ECON	VERDICT
(a) 8×H100 + prefix cache	✓	✓	✓	✗ capacity — ~5 hosts
(b) P/D 2-pool	✓	✓	~2×	✗ at 1 prefill host → scales to ~5
(c) KV-quant 7B	✓	cheap	✓	✗ quality — excluded



⑤ OUTCOME — SURVIVORS

(a) latency baseline (b) prefill pool scales to ~5

✗ REJECTED

(c) fails quality gate